

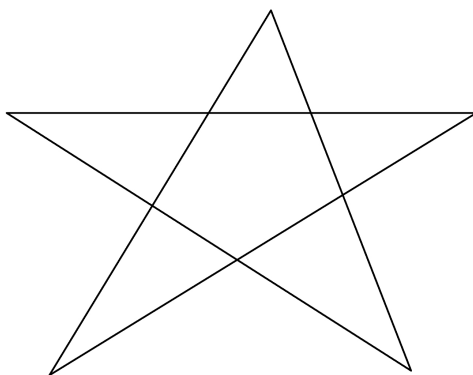
## Problem Set #22

UMass Recreational Math Club

May 2025

**Problem 1.** Given  $x^{x^{x^{\dots}}} = 2$ , solve for  $x$ .

**Problem 2.** Consider a pentagram (see figure below). There are five disjoint triangles initially, only counting the five “caps” of the pentagram (do not count the larger triangles formed by using three of the five vertices, as those have lines going through them). By adding two lines, you can go from having five to ten disjoint triangles. How?



**Problem 3.** 100 people are waiting to board a plane. The first person’s ticket says Seat 1; the second person in line has a ticket that says Seat 2, and so on. The first person ignores the fact that their ticket says Seat 1, and randomly chooses one of the hundred seats. From this point on, the next 98 people will always sit in their assigned seats if possible; if their seat is taken, they will randomly choose one of the remaining seats (after the first person, the second person takes a seat; after the second person, the third person takes a seat, and so on). What is the probability the 100th person sits in Seat 100?

**Problem 4.** You have 12 coins, one weighs slightly less or slightly more than the others. Using an equal arm balance and only making three weighings determine which one is different and whether it is slightly less or slightly more. Warning: the fact that you don’t know whether it is less or more is a major problem in the solution.